جامعة الحسن الثاني بالدار البيضاء
UNIVERSITÉ HASSAN II DE CASABLANCA

ECOLE NORMALE SUPERIEURE DE L'ENSEIGNEMENT TECHNIQUE DE MOHAMMEDIA

ENSET

DÉPARTEMENT MATHÉMATIQUE INFORMATIQUE

Architectures distribuées et Middlewares

Rapport du contrôle

application de géstion des contrats d'assurance

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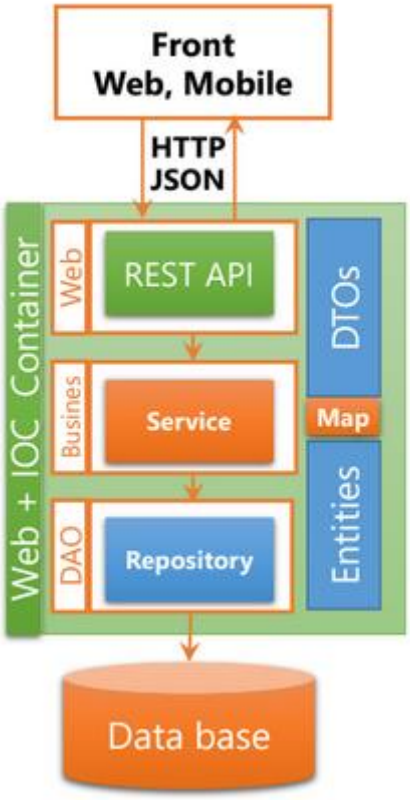
Introduction

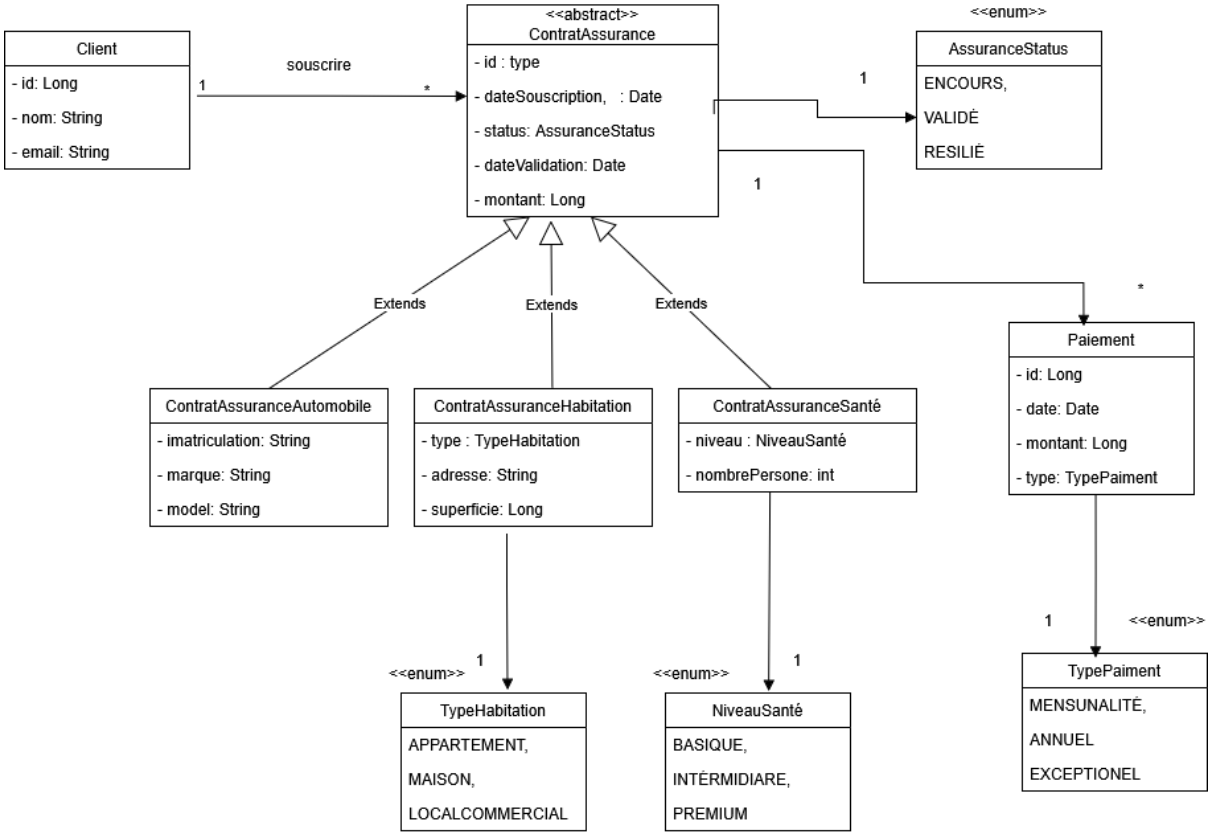
On souhaite concevoir et développer une application Web JEE basée sur le Framework Spring et Angular. L’application permet de gérer des contrats d’assurance.

Conception :

1-architecture technique du projet

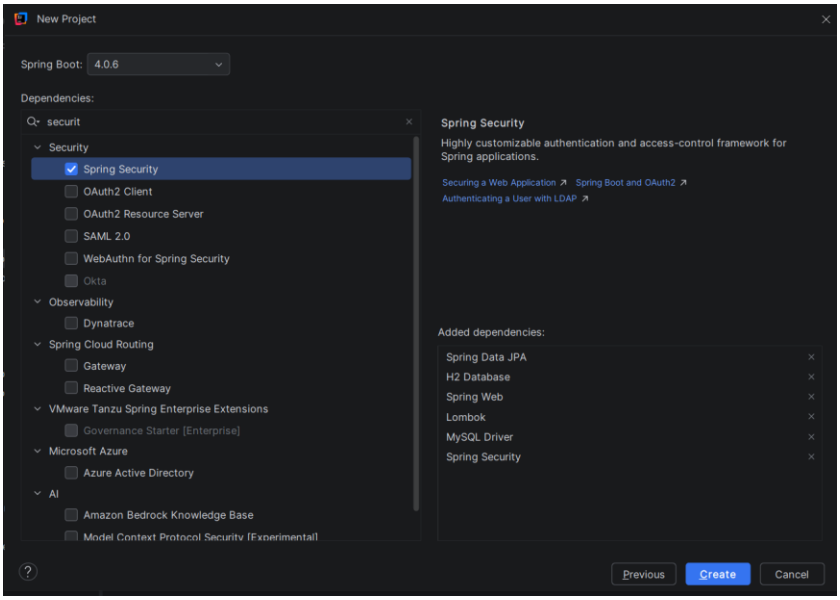


2-diagramme de classes

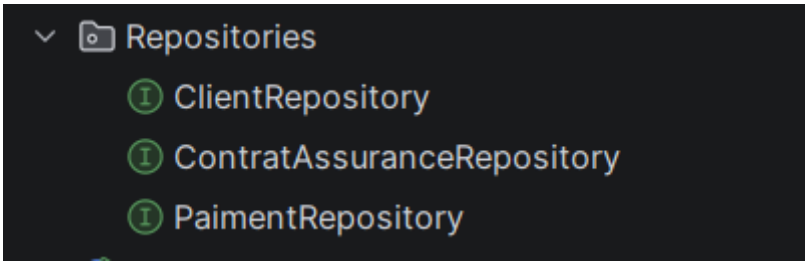


Implémentation

1-un projet Spring boot avec les dépendances requises



2-Couche DAO

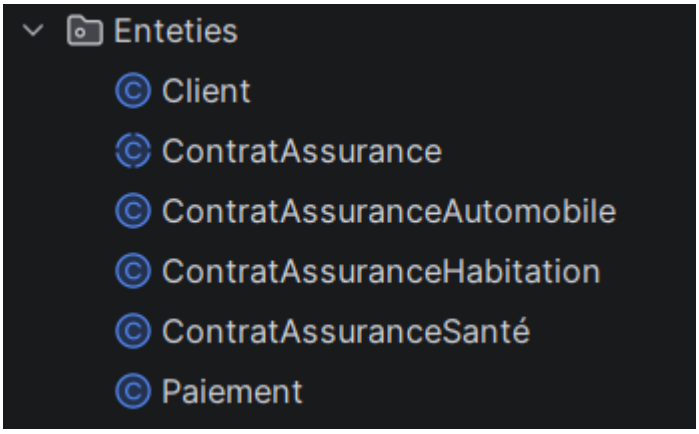



```
package net.hamza.gestiondecontractsassurance.Repositories;

import net.hamza.gestiondecontractsassurance.Enteties.Client;
import org.springframework.data.jpa.repository.JpaRepository;

public interface ClientRepository extends JpaRepository<Client,Long> {
}
```

Entities



Test de la couche DAO

```
@Bean
CommandLineRunner start(
        ClientRepository clientRepository,
        ContratAssuranceRepository contratAssuranceRepository,
        PaimentRepository paimentRepository
) {
    return args -> {

        ClientMapper clientMapper = new ClientMapper();

        Stream.of("name1", "name2", "name3").forEach(name -> {

            // Save Client
            ClientDto clientDto = new ClientDto();
            clientDto.setNom(name);
            clientDto.setEmail(name + "@gmail.com");

            Client client = clientRepository.save(
                    clientMapper.fromClientDto(clientDto)
```

```
        );

        // Contrat Assurance Habitation
        for (int i = 0; i < 3; i++) {

            ContratAssuranceHabitation cah =
                    new ContratAssuranceHabitation();

            cah.setDateSouscription(new Date());
            cah.setDateValidation(new Date());
            cah.setAdresse("Adresse " + i);
            cah.setSuperficie(120L + i);
            cah.setMontant(10000L + i * 1000);
            cah.setStatus(AssuranceStatus.ENCOURS);
            cah.setType(TypeHabitation.APPARTEMENT);
            cah.setClient(client);

            contratAssuranceRepository.save(cah);
        }

        // Contrat Assurance Automobile
        for (int i = 0; i < 3; i++) {

            ContratAssuranceAutomobile caa =
                    new ContratAssuranceAutomobile();

            caa.setDateSouscription(new Date());
            caa.setDateValidation(new Date());
            caa.setImatriculation("12345-A-" + i);
            caa.setMarque("BMW");
            caa.setModel("X" + i);
            caa.setMontant(20000L + i * 1000);
            caa.setStatus(AssuranceStatus.ENCOURS);
            caa.setClient(client);

            contratAssuranceRepository.save(caa);
        }

        // Contrat Assurance Santé
        for (int i = 0; i < 3; i++) {

            ContratAssuranceSanté cas =
                    new ContratAssuranceSanté();

            cas.setDateSouscription(new Date());
            cas.setDateValidation(new Date());
            cas.setNombrePersonne(2 + i);
            cas.setNiveau(NiveauSanté.PREMIUM);
            cas.setMontant(15000L + i * 1000);
            cas.setStatus(AssuranceStatus.ENCOURS);
            cas.setClient(client);

            contratAssuranceRepository.save(cas);
        }
    });

    // Paiements
    contratAssuranceRepository.findAll().forEach(contrat -> {

        for (int i = 0; i < 2; i++) {
```

```
                Paiement paiement = new Paiement();

                paiement.setDate(new Date());
                paiement.setMontant(5000L + i * 1000);
                paiement.setType(TypePaiment.MENSUNALITÉ);

                paiement.setContratAssurance(contrat);

                paimentRepository.save(paiement);
            }
        });
    };
```

			id	email	nom
Edit	Copy	Delete	1	name1@gmail.com	name1
Edit	Copy	Delete	2	name2@gmail.com	name2
Edit	Copy	Delete	3	name3@gmail.com	name3

			nombre_personne	typeassurance	client_id	date_souscription	date_validation	id	montant	superficie	adresse	imatriculation	mar
Edit	Copy	Delete	NULL	CAH	1	2026-05-11 12:03:06.000000	2026-05-11 12:03:06.000000	1	10000	120	Adresse 0	NULL	NUL
Edit	Copy	Delete	NULL	CAH	1	2026-05-11 12:03:06.000000	2026-05-11 12:03:06.000000	2	11000	121	Adresse 1	NULL	NUL
Edit	Copy	Delete	NULL	CAH	1	2026-05-11 12:03:06.000000	2026-05-11 12:03:06.000000	3	12000	122	Adresse 2	NULL	NUL
Edit	Copy	Delete	NULL	CAA	1	2026-05-11 12:03:07.000000	2026-05-11 12:03:07.000000	4	20000	NULL	NULL	12345-A-0	BMV
Edit	Copy	Delete	NULL	CAA	1	2026-05-11 12:03:07.000000	2026-05-11 12:03:07.000000	5	21000	NULL	NULL	12345-A-1	BMV
Edit	Copy	Delete	NULL	CAA	1	2026-05-11 12:03:07.000000	2026-05-11 12:03:07.000000	6	22000	NULL	NULL	12345-A-2	BMV
Edit	Copy	Delete	2	CAS	1	2026-05-11 12:03:07.000000	2026-05-11 12:03:07.000000	7	15000	NULL	NULL	NULL	NUL
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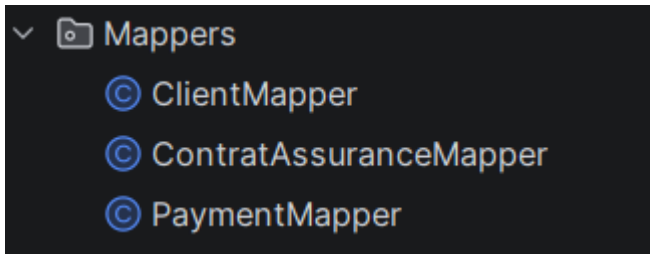
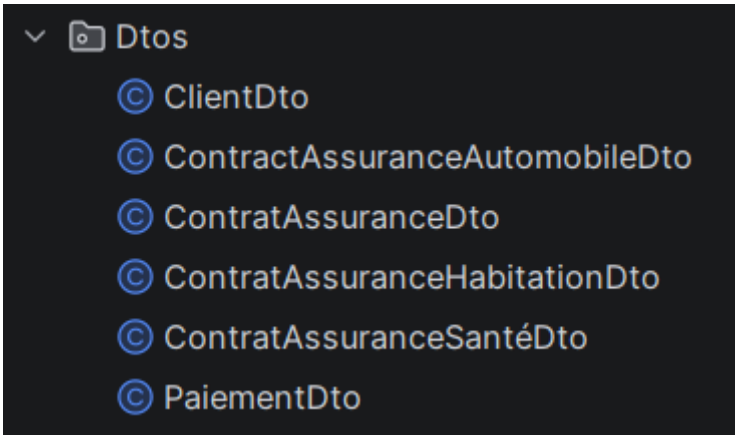
Edit	Copy	Delete	NULL	CAH	2	2026-05-11 12:03:07.000000	2026-05-11 12:03:07.000000	10	10000	120	Adresse 0	NULL	NUL

			contrat_assurance_id	date	id	montant	type
Edit	Copy	Delete	1	2026-05-11 12:03:08.000000	1	5000	MENSUNALITÉ
Edit	Copy	Delete	1	2026-05-11 12:03:08.000000	2	6000	MENSUNALITÉ
Edit	Copy	Delete	2	2026-05-11 12:03:08.000000	3	5000	MENSUNALITÉ
Edit	Copy	Delete	2	2026-05-11 12:03:08.000000	4	6000	MENSUNALITÉ
Edit	Copy	Delete	3	2026-05-11 12:03:08.000000	5	5000	MENSUNALITÉ
Edit	Copy	Delete	3	2026-05-11 12:03:08.000000	6	6000	MENSUNALITÉ
Edit	Copy	Delete	4	2026-05-11 12:03:08.000000	7	5000	MENSUNALITÉ
Edit	Copy	Delete	4	2026-05-11 12:03:08.000000	8	6000	MENSUNALITÉ
Edit	Copy	Delete	5	2026-05-11 12:03:08.000000	9	5000	MENSUNALITÉ
Edit	Copy	Delete	5	2026-05-11 12:03:08.000000	10	6000	MENSUNALITÉ
Edit	Copy	Delete	6	2026-05-11 12:03:08.000000	11	5000	MENSUNALITÉ
Edit	Copy	Delete	6	2026-05-11 12:03:08.000000	12	6000	MENSUNALITÉ
Edit	Copy	Delete	7	2026-05-11 12:03:08.000000	13	5000	MENSUNALITÉ
Edit	Copy	Delete	7	2026-05-11 12:03:08.000000	14	6000	MENSUNALITÉ

3-es DTOs et les Mappers